

Model Equations

Common conventions and units [1]

All three multicomponent models are implemented in dimensionless form. Throughout this document,

$$D_i \equiv \partial_i - iA_i, \quad D_t \equiv \partial_t + i\mu, \quad i \in \{x, y\}, \quad (1)$$

where μ is the scalar electric potential (the same quantity is sometimes denoted ϕ). Complex conjugation is denoted by an asterisk. The gauge transformation

$$\psi_n \rightarrow e^{i\chi} \psi_n, \quad \mathbf{A} \rightarrow \mathbf{A} + \nabla \chi, \quad \mu \rightarrow \mu - \partial_t \chi \quad (2)$$

leaves all of the equations below invariant.

The pyTDGL dimensional scales, in SI, are [1]

$$\tau_0 = \mu_0 \sigma \lambda^2, \quad B_0 = B_{c2} = \frac{\Phi_0}{2\pi \xi^2}, \quad A_0 = \xi B_{c2} = \frac{\Phi_0}{2\pi \xi}, \quad (3)$$

$$J_0 = \frac{4\xi B_{c2}}{\mu_0 \lambda^2}, \quad K_0 = J_0 d = \frac{4\xi B_{c2}}{\mu_0 \Lambda}, \quad V_0 = \frac{\xi J_0}{\sigma} = \frac{4\xi^2 B_{c2}}{\mu_0 \sigma \lambda^2}, \quad (4)$$

where ξ is the coherence length, λ the London penetration depth, d the film thickness, $\Lambda = \lambda^2/d$, σ the normal-state conductivity, and $\Phi_0 = h/(2e)$ the flux quantum. For the single-band, $d + d'$, and $s + s$ models, solver variables are

$$\mathbf{r} = \frac{\mathbf{r}_{\text{phys}}}{\xi}, \quad t = \frac{t_{\text{phys}}}{\tau_0}, \quad \mathbf{A} = \frac{\mathbf{A}_{\text{phys}}}{A_0}, \quad \mathbf{B} = \frac{\mathbf{B}_{\text{phys}}}{B_0}, \quad \mu = \frac{V_{\text{phys}}}{V_0}, \quad \mathbf{j} = \frac{\mathbf{K}_{\text{phys}}}{K_0}. \quad (5)$$

Order parameters and code coefficients are dimensionless. The source papers use Gaussian cgs; all dimensional formulas below are converted to SI, while dimensionless ratios are unit-system independent.

The $s + d$ equations retain the source paper's order-parameter diffusion clock $t_\psi = \xi_d^2/\mathcal{D}$. Their numerical transport current is therefore $\mathbf{J}_{\text{cond}}/\beta_{\text{em}}$. Terminal currents are divided by β_{em} on input and solution currents are multiplied by it on output, so public currents still use the physical scale K_0 in Eq. (4).

The $s + s$ equations use two current conversions. First, $c_{\text{em}} \equiv \text{em_coupling}$ maps the variational condensate current to the model's transport convention. Second, $\beta_{\text{em}} \equiv \text{beta_em}$ sets the electromagnetic clock. The current stored by the solver and used by the Poisson equation is $\mathbf{j}_s = \mathbf{J}_{\text{cond}}/(c_{\text{em}}\beta_{\text{em}})$. Terminal currents are divided by β_{em} on input and solution currents are multiplied by it on output, so the new coefficient is backward compatible at its default value one.

In the legacy arbitrary-mesh backend, current conservation is imposed through

$$\mathbf{j}_n = -\nabla \mu - \partial_t \mathbf{A}, \quad (6)$$

$$\nabla^2 \mu = \nabla \cdot \mathbf{j}_s - \nabla \cdot \partial_t \mathbf{A}, \quad (7)$$

where $\mathbf{j}_s$ is the Poisson-normalized supercurrent specified for each model below. The $d + d'$ model uses a prescribed vector potential. The $s + d$ and $s + s$ models can also implement the local electromagnetic evolution given in their respective sections below. The standalone magnetic-periodic mode instead uses the source paper's temporal gauge $\mu = 0$ and does not invoke the arbitrary-mesh scalar-potential solve.

1 Magnetic-periodic rectangular operator mode

The magnetic-periodic mode is a standalone structured-grid backend for a rectangular vortex cell $\Omega = [0, L_x) \times [0, L_y)$. It is distinct from the legacy arbitrary-mesh backend: it neither adds seam edges to an open mesh nor applies physical boundary conditions on the perimeter of a finite sample. A cell with L_x and L_y expressed in coherence-length units and $n \in \mathbb{Z}$ flux quanta has the fixed mean induction

$$\bar{B} = \frac{2\pi n}{L_x L_y}. \quad (8)$$

In the Landau gauge $\bar{\mathbf{A}} = (0, \bar{B}x)$, both condensates ψ_α , $\alpha \in \{1, 2\}$, obey the same magnetic translations,

$$\psi_\alpha(x + L_x, y) = e^{i\bar{B}L_x y} \psi_\alpha(x, y), \quad \psi_\alpha(x, y + L_y) = \psi_\alpha(x, y). \quad (9)$$

Equation (8) makes the two translations consistent at a corner because $e^{i\bar{B}L_x L_y} = e^{i2\pi n} = 1$.

The evolved field is a periodic vector potential $\mathbf{a}$,

$$\mathbf{A} = \bar{\mathbf{A}} + \mathbf{a}, \quad \mathbf{a}(x + L_x, y) = \mathbf{a}(x, y), \quad \mathbf{a}(x, y + L_y) = \mathbf{a}(x, y), \quad B = \bar{B} + (\nabla \times \mathbf{a})_z. \quad (10)$$

Consequently,

$$\int_{\Omega} B d^2r = \bar{B} L_x L_y = 2\pi n, \quad (11)$$

since the integral of the curl of a periodic field vanishes. There is no physical perimeter and hence no vacuum condition $B = H$. An external H may appear in a Gibbs-energy diagnostic through $\kappa^2(B - H)^2$, but it does not replace the fixed-flux constraint in the magnetic-periodic evolution. For a stationary nonzero-flux cell with spatially uniform GL coefficients, the applied field can be recovered in postprocessing from the virial relation [4],

$$H_{\text{vir}} = \frac{\langle f_{\text{grad}} \rangle + 2\kappa^2 \langle B^2 \rangle}{2\kappa^2 \bar{B}}, \quad \bar{B} \neq 0, \quad (12)$$

where f_{grad} includes both diagonal condensate stiffnesses and the mixed-gradient term. This form is not used for a spatially fixed disorder profile, whose scaling adds an inhomogeneity contribution to the general virial theorem.

On an $N_x \times N_y$ Cartesian grid, let U_{ij}^x and U_{ij}^y be the links of $\bar{\mathbf{A}} + \mathbf{a}$. The structured covariant forward differences are

$$(D_x^h \psi)_{ij} = \frac{U_{ij}^x \psi_{i+1 \bmod N_x, j} - \psi_{ij}}{h_x}, \quad (13)$$

$$(D_y^h \psi)_{ij} = \frac{U_{ij}^y \psi_{i, j+1 \bmod N_y} - \psi_{ij}}{h_y}. \quad (14)$$

The wrap links contain the phases in Eq. (9). If C is the oriented bond-to-plaquette curl and C^T its energy adjoint, then

$$\mathbf{B} = \bar{B} \mathbf{1} + C \mathbf{a}, \quad \beta_{\text{em}} \dot{\mathbf{a}} = \mathbf{J}_{\text{cond}} - \mathbf{J}_{\text{drive}} - \kappa^2 C^T C \mathbf{a}. \quad (15)$$

Thus the magnetic operator is the periodic positive-semidefinite curl–curl operator $C^T C$ and cannot change the mean flux. By contrast, the legacy arbitrary-mesh operators retain their variational order-parameter boundary conditions and, for local screening, the physical-perimeter condition

$B = H$. The two uniform harmonic link modes are retained. A bulk drive can therefore advance the unwrapped uniform part of $\mathbf{a}$ and generate a nonzero mean $\mathbf{E} = -\partial_t \mathbf{a}$ without changing the flux sector. The initial standalone implementation is exposed through `MagneticPeriodicCell`, `MagneticPeriodicSolver`, `MagneticPeriodicSolution`, and `solve_magnetic_periodic`. It supports the $s + d$ model on the CPU with a static fixed-flux cell and local $\phi = 0$ dynamics.

2 s+d [2, 5, 3]

2.1 Order-parameter equations

Let ψ_d and ψ_s denote the dimensionless d - and s -wave order parameters. The implemented mixed-symmetry TDGL equations follow the dimensionless $d + s$ framework of Refs. [2, 5]:

$$D_t \psi_d = (D_x^2 + D_y^2) \psi_d + \varepsilon \psi_d - |\psi_d|^2 \psi_d - \frac{\tau_3}{2} |\psi_s|^2 \psi_d - \tau_4 \psi_s^2 \psi_d^* + \eta_v (D_y^2 - D_x^2) \psi_s, \quad (16)$$

$$\eta_s r_s D_t \psi_s = \eta_s (D_x^2 + D_y^2) \psi_s + \nu_{\text{eff}} \psi_s - \tau_1 |\psi_s|^2 \psi_s - \frac{\tau_3}{2} |\psi_d|^2 \psi_s - \tau_4 \psi_d^2 \psi_s^* + \eta_v (D_y^2 - D_x^2) \psi_d. \quad (17)$$

Here $\varepsilon = \varepsilon(\mathbf{r}, t)$ is the solver input `disorder_epsilon`. The effective s -sector coefficient is

$$\nu_{\text{eff}} = \nu + c_\nu (\varepsilon - 1), \quad c_\nu = \text{nu_disorder_coupling}. \quad (18)$$

Thus ε changes only the d quadratic drive when $c_\nu = 0$, which is the default. A nonzero c_ν allows the same spatial profile to model correlated suppression of both components, as needed for the twin boundaries in Ref. [3]. The standard homogeneous value is $\varepsilon = 1$. The dimensionless d -sector stiffness and kinetic coefficient are both one: Gonçalves *et al.* use the physical d -band stiffness to define ξ_d and the common diffusion clock $t_0 = \xi_d^2 / \mathcal{D}$. Thus the d relaxation scale is implicit rather than absent. The optional positive factor r_s (`relaxation_s`) multiplies only the s -sector kinetic coefficient. Setting $r_s = 1$ recovers Gonçalves Eqs. (17)–(18); allowing $r_s \neq 1$ represents a relative s -band relaxation time that is independent of the stiffness ratio η_s . The legacy `Layer.u` coefficient is used only by the single-band KWT equation and does not set either relaxation scale in this multicomponent model.

Define the mixed directional currents

$$Q_i = \text{Im}(\psi_d^* D_i \psi_s + \psi_s^* D_i \psi_d). \quad (19)$$

The condensate current used by the implementation before electromagnetic normalization is

$$J_{\text{cond},x} = \text{Im}(\psi_d^* D_x \psi_d) + \eta_s \text{Im}(\psi_s^* D_x \psi_s) - \eta_v Q_x, \quad (20)$$

$$J_{\text{cond},y} = \text{Im}(\psi_d^* D_y \psi_d) + \eta_s \text{Im}(\psi_s^* D_y \psi_s) + \eta_v Q_y. \quad (21)$$

The transport-normalized current stored by the solver and used in Eq. (7) is

$$\mathbf{j}_s = \frac{\mathbf{J}_{\text{cond}}}{\beta_{\text{em}}}. \quad (22)$$

The opposite signs of the x and y mixed-current terms encode the $d_{x^2-y^2}$ symmetry.

The implemented diagnostic `compute_s_plus_d_free_energy` evaluates the discrete counterpart of the Gibbs functional

$$\mathcal{G} = \frac{1}{|\Omega|} \int_{\Omega} d^2 r \left[-\varepsilon |\psi_d|^2 - \nu_{\text{eff}} |\psi_s|^2 + \frac{1}{2} |\psi_d|^4 + \frac{\tau_1}{2} |\psi_s|^4 + \frac{\tau_3}{2} |\psi_d|^2 |\psi_s|^2 \right. \quad (23)$$

$$\left. + \frac{\tau_4}{2} (\psi_s^{*2} \psi_d^2 + \psi_d^{*2} \psi_s^2) + |\mathbf{D} \psi_d|^2 + \eta_s |\mathbf{D} \psi_s|^2 \right. \quad (24)$$

$$\left. + \eta_v \{ (D_y \psi_s)^* D_y \psi_d + \text{c.c.} - (D_x \psi_s)^* D_x \psi_d - \text{c.c.} \} + \kappa_d^2 (B - H)^2 \right]. \quad (25)$$

For the homogeneous paper model, $\varepsilon = 1$; this is Eqs. (78)–(79) with the same gauge and normalization conventions as above.

With `include_screening=True`, the vector potential obeys

$$\beta_{\text{em}} \partial_t \mathbf{A} = \mathbf{J}_{\text{cond}} - \kappa_d^2 (\partial_y B, -\partial_x B), \quad B = (\nabla \times \mathbf{A})_z, \quad (26)$$

which is Eq. (19) of Gonçalves *et al.* in the $\phi = 0$ gauge. The legacy arbitrary-mesh implementation writes $\mathbf{A} = \mathbf{A}_{\text{applied}} + \mathbf{A}_{\text{induced}}$ and applies magnetic diffusion to $B - H = \nabla \times \mathbf{A}_{\text{induced}}$. This is equivalent to the paper’s $B = H$ vacuum boundary condition for a uniform applied field and remains variationally consistent for a nonuniform prescribed field.

For an insulating boundary of the legacy arbitrary-mesh backend, with outward normal $\mathbf{n}$, the natural order-parameter conditions generated by the same gradient functional are

$$\mathbf{n} \cdot \mathbf{D} \psi_d + \eta_v (-n_x D_x + n_y D_y) \psi_s = 0, \quad (27)$$

$$\eta_s \mathbf{n} \cdot \mathbf{D} \psi_s + \eta_v (-n_x D_x + n_y D_y) \psi_d = 0. \quad (28)$$

Substitution into the condensate current of Eq. (8) gives $\mathbf{J}_s \cdot \mathbf{n} = 0$, precisely Eq. (12) of Gonçalves *et al.*; the finite-element assembly enforces these as variational natural boundary conditions.

2.2 Dimensional origin and SI units

From effective masses m_d^*, m_s^*, m_v^* , quadratic coefficients α_d, α_s , quartic coefficients $\beta_1, \dots, \beta_4$, diffusion coefficient $\mathcal{D}$, and conductivity σ , Gonçalves *et al.* obtain [2]

$$\eta_s = \frac{m_d^*}{m_s^*}, \quad \eta_v = \frac{m_d^*}{m_v^*}, \quad \nu = \frac{\alpha_s}{\alpha_d}, \quad (29)$$

$$\tau_1 = \frac{\beta_1}{\beta_2}, \quad \tau_3 = \frac{\beta_3}{\beta_2}, \quad \tau_4 = \frac{\beta_4}{\beta_2}, \quad (30)$$

$$\beta = \mu_0 \sigma \kappa_d^2 \mathcal{D} = \frac{\mu_0 \sigma \lambda_d^2}{\xi_d^2 / \mathcal{D}}. \quad (31)$$

β is the ratio of the electromagnetic relaxation time $\mu_0 \sigma \lambda_d^2$ to the order-parameter diffusion time $\xi_d^2 / \mathcal{D}$; Eq. (31) is the SI conversion of the paper’s expression.

The remaining source scales are

$$|\psi_{d0}|^2 = \frac{\alpha_d}{2\beta_2}, \quad \xi_d^2 = \frac{\hbar^2}{2m_d^* \alpha_d}, \quad \lambda_d^2 = \frac{2m_d^* \beta_2}{\mu_0 (e^*)^2 \alpha_d}, \quad (32)$$

$$\kappa_d = \frac{\lambda_d}{\xi_d}, \quad B_{0,d} \equiv H_{c2,d}^{(0)} = \frac{\Phi_0}{2\pi \xi_d^2}, \quad \Phi_0 = \frac{h}{2e}. \quad (33)$$

SI units are: m_i^* [kg], $\mathcal{D}$ [$\text{m}^2 \text{s}^{-1}$], σ [S m^{-1}], ξ_d, λ_d [m], B [T], $\mathbf{A}$ [T m], and Φ_0 [Wb]. Each set (α_s, α_d) and $(\beta_1, \dots, \beta_4)$ must use common compatible units; their ratios are dimensionless for any consistent order-parameter normalization. $B_{0,d}$ is the nominal pure- d field scale stored by the code as `device.Bc2` and denoted B_0 by Li *et al.*; it is not the α_s -dependent upper critical field of the mixed state.

2.3 Mapping the Li–Wang–Wang dynamics

Li, Wang, and Wang write separate phenomenological kinetic coefficients η_s^L and η_d^L for their fields S and D , with $q \equiv \eta_s^L = 2\eta_d^L$ [3]. Identifying $\psi_s = S$ and $\psi_d = D$, first multiply their S equation

by two so that its mixed-gradient term has the same variational coefficient as the D equation. The static parameter map is then

$$\eta_s = 2, \quad \eta_v = -1, \quad \nu = 2\alpha_s, \quad \tau_1 = \frac{8}{3}, \quad \tau_3 = \frac{16}{3}, \quad \tau_4 = \frac{4}{3}. \quad (34)$$

Use the d -component clock

$$t_{\text{code}} = \frac{t_L}{\eta_d^L} = \frac{2t_L}{q}. \quad (35)$$

The D kinetic coefficient is thereby one, while the S coefficient is $2\eta_s^L/\eta_d^L = 4$. Since the code writes that coefficient as $\eta_s r_s$, the dynamical map is

$$r_s = 2, \quad \beta_{\text{em}} = \frac{1}{\eta_d^L} = \frac{2}{q}, \quad \mathbf{E}_L = -\partial_{t_L} \mathbf{A} = \beta_{\text{em}} \mathbf{E}_{\text{code}}. \quad (36)$$

Consequently, q changes the common physical clock and electromagnetic coefficient, whereas $r_s = 2$ carries the relative S – D kinetic factor that cannot be represented by the static stiffness η_s alone.

For a twin profile $g(\mathbf{r})$ with $\alpha_i = \alpha_{i,0} - u_i g$, the script uses

$$\varepsilon = 1 - u_d g, \quad c_\nu = \frac{2u_s}{u_d}, \quad (37)$$

so that Eq. (18) gives $\nu_{\text{eff}} = 2\alpha_s - 2u_s g$. The ratio is used only for $u_d > 0$.

2.4 Parameters

N.B. Theoretical values for the parameters have been derived microscopically, corresponding to those in section VIII of Goncalves, but we allow them to be adjusted just in case.

Code name	Meaning	Default
η_s (eta_s)	Effective-mass ratio m_d^*/m_s^* ; sets the relative s -sector stiffness and the base kinetic multiplier. The full coefficient of $D_t \psi_s$ is $\eta_s r_s$.	1
r_s (relaxation_s)	Extra s -sector kinetic multiplier; $r_s = 1$ is the Gonçalves common-diffusion model, while $\eta_s r_s$ is the implemented s coefficient of $D_t \psi_s$.	1
η_v (eta_v)	Mixed-gradient effective-mass ratio m_d^*/m_v^* , responsible for the anisotropic s – d coupling.	0
ν (nu)	Homogeneous quadratic ratio α_s/α_d ; it is the base value in the free-energy term $-\nu_{\text{eff}} \psi_s ^2$.	–1
c_ν (nu_disorder_coupling)	Optional coupling of the disorder profile into the s quadratic coefficient, $\nu_{\text{eff}} = \nu + c_\nu(\varepsilon - 1)$.	0
τ_1 (tau1)	Quartic ratio β_1/β_2 , comparing the s self-interaction with the d self-interaction.	1
τ_3 (tau3)	Density-coupling ratio β_3/β_2 ; TDGL cross-density coefficient $\tau_3/2$.	0
τ_4 (tau4)	Phase-sensitive quartic ratio β_4/β_2 , coupling $\psi_s^2 \psi_d^*$ and $\psi_d^2 \psi_s^*$.	0
β_{em} (beta_em)	SI electromagnetic factor $\beta = \mu_0 \sigma \kappa_d^2 \mathcal{D}$; it controls Eq. (19) when screening is enabled and divides the stored transport current in Eq. (22).	1

Code name	Meaning	Default
ε (<code>disorder_epsilon</code>)	Local d -sector quadratic coefficient, interpretable as $\alpha_d(\mathbf{r}, t)/\alpha_{d,\text{ref}}$; the homogeneous Gonçalves normalization gives $\varepsilon = 1$.	1

The validation conditions are

$$\eta_s > 0, \quad r_s > 0, \quad \eta_s > \eta_v^2, \quad \tau_1 > 0, \quad \beta_{\text{em}} > 0, \quad (38)$$

with $\varepsilon \leq 1$ enforced by the solver. The condition $\eta_s > \eta_v^2$ makes the mixed-gradient sector positive definite.

2.5 Limitations and approximations

The Gonçalves formulation treats the diffusion coefficient and conductivity phenomenologically and does not impose a strict clean- or dirty-limit condition. The microscopic theory of Zhu *et al.* includes magnetic and nonmagnetic impurity scattering rather than assuming only the asymptotic dirty limit. Nevertheless, the implementation's local Ohmic normal current and diffusive relaxation are dirty-limit-like; quantitative dynamics require material-specific scattering and relaxation parameters.

By default the field $\varepsilon(\mathbf{r}, t)$ changes only the d -sector quadratic coefficient; Eq. (18) optionally correlates the s -sector suppression. This remains a phenomenological rather than a complete microscopic disorder model. A scalar terminal value fixes ψ_d and sets $\psi_s = 0$ at terminal sites, which is a numerical contact model rather than a general $s + d$ interface condition.

The nominal $H_{c2,d}^{(0)}$ used for nondimensionalization is the pure- d scale. The mixed-state upper critical field is generally different and must be computed from the coupled equations, for example from the linearized Li eigenproblem used by the reproduction script.

3 d+d' [6, 7]

3.1 Free energy and equilibrium-relaxation equations

Let d denote the dominant $d_{x^2-y^2}$ component and d' the subdominant d_{xy} component. In the code they are stored as $\psi_1 = d$ and $\psi_2 = d'$, respectively. The dimensionless free-energy density of Lei, Aruna, and Wang [6] is augmented by the optional orbital-Zeeman coupling derived microscopically by Wang and Wang [7]:

$$\begin{aligned} f_{d+d'} = & -|d|^2 - \alpha|d'|^2 + \frac{1}{2}(|d|^4 + |d'|^4) + \frac{1}{3}|d|^2|d'|^2 + \frac{1}{6}(d^*d' + dd'^*)^2 \\ & + |\mathbf{D}d|^2 + |\mathbf{D}d'|^2 + \kappa^2 b^2 - ig_Z b (d^*d' - dd'^*), \end{aligned} \quad (39)$$

where

$$b = (\nabla \times \mathbf{A})_z = \frac{B_z}{B_{c2}}, \quad B_{c2} = \frac{\Phi_0}{2\pi\xi^2}. \quad (40)$$

The original field-driven model in Ref. [6] deliberately omits the last term; setting $g_Z = 0$ reproduces that functional. The signed coefficient g_Z incorporates the microscopic particle-hole asymmetry and the dimensionless normalization of the orbital-Zeeman interaction. The code computes b at mesh sites by a discrete Stokes curl of the edge-centered applied vector potential.

The source papers specify a static GL functional rather than physical TDGL kinetics. The implemented equations are therefore phenomenological negative gradient flow for finding stationary

states:

$$r_d D_t d = (D_x^2 + D_y^2) d + d - |d|^2 d - \frac{2}{3} |d'|^2 d - \frac{1}{3} d'^2 d^* + i g_Z b d', \quad (41)$$

$$r_{d'} D_t d' = (D_x^2 + D_y^2) d' + \alpha d' - |d'|^2 d' - \frac{2}{3} |d|^2 d' - \frac{1}{3} d^2 d'^* - i g_Z b d. \quad (42)$$

The factors 2/3 and 1/3 follow by differentiating both quartic cross terms in Eq. (39); they are not independent fitting parameters.

At $b = 0$ and $g_Z = 0$, a uniform pure- d stationary state has $|d| = 1$ and $d' = 0$. For $1/3 < \alpha < 1$, the uniform mixed minimum instead has

$$|d|^2 = \frac{3(3-\alpha)}{8}, \quad |d'|^2 = \frac{3(3\alpha-1)}{8}, \quad \arg(d') - \arg(d) = \pm \frac{\pi}{2}. \quad (43)$$

For $g_Z b > 0$, the Zeeman term lifts this chiral degeneracy and favors $d'/d = -i|d'|/|d|$; reversing the field reverses the favored chirality.

3.2 Current and orbital magnetization

Because Eq. (39) has equal diagonal stiffnesses and no mixed-gradient term, the implemented condensate and transport currents are

$$\mathbf{J}_{\text{diag}} = \text{Im}(d^* \mathbf{D} d) + \text{Im}(d'^* \mathbf{D} d'), \quad (44)$$

$$\mathbf{J}_{\text{cond}} = \mathbf{J}_{\text{diag}} + \mathbf{J}_Z, \quad (45)$$

$$\mathbf{j}_s = \frac{\mathbf{J}_{\text{cond}}}{c_{\text{em}}}, \quad c_{\text{em}} \equiv \text{em.coupling}. \quad (46)$$

The local dimensionless orbital magnetization associated with the Zeeman term is

$$m_z = - \left. \frac{\partial f_Z}{\partial b} \right|_{d,d'} = i g_Z (d^* d' - d d'^*), \quad f_Z = -i g_Z b (d^* d' - d d'^*), \quad (47)$$

This is the order-parameter orbital contribution and excludes the derivative of the separately prescribed $\kappa^2 b^2$ field energy. Equation (47) is exposed as the diagnostic `Solution.orbital_magnetization`. The bound current $\mathbf{J}_Z$ is evaluated as the exact discrete derivative of $-\int m_z b d^2 r$ with respect to the edge vector potential. It is included in the current used by the transport Poisson solve.

The discrete condensate free-energy diagnostic includes the local potential, gradient energy, and orbital-Zeeman energy in Eq. (39). It omits $\kappa^2 b^2$, which is constant with respect to d, d' when $\mathbf{A}$ is prescribed and therefore does not affect order-parameter relaxation.

3.3 Parameters

Code name	Meaning	Default
α (<code>alpha</code>)	Ratio of the subdominant and dominant quadratic GL coefficients in the normalization of Ref. [6]; controls the zero-field d_{xy} instability.	0.5
r_d (<code>relaxation.d</code>)	Positive phenomenological relaxation coefficient for the dominant $d_{x^2-y^2}$ component.	1

Code name	Meaning	Default
$r_{d'}$ (<code>relaxation_d_prime</code>)	Positive phenomenological relaxation coefficient for the subdominant d_{xy} component.	1
c_{em} (<code>em_coupling</code>)	Implementation-specific conversion from condensate-current normalization to the transport normalization used by the Poisson solver.	1
g_Z (<code>zeeman_coupling</code>)	Signed dimensionless orbital-Zeeman coefficient; zero selects the unmodified Lei–Aruna–Wang model.	0

All parameters must be finite, $r_d, r_{d'}, c_{\text{em}} > 0$, and the current implementation requires $|g_Z| < 1$. Negative α is permitted. Values $\alpha > 1$ are numerically allowed with a warning because they leave the assumed subdominant- d_{xy} regime of the source model.

3.4 Limitations and approximations

Our equations are weak-coupling microscopic results based on BCS, Gor'kov, and path-integral methods without impurity scattering; they do not assume the dirty limit. The coefficients $r_d, r_{d'}$ are phenomenological additions, so the simulated trajectory is equilibrium relaxation rather than a microscopic clean-limit TDGL prediction.

The Zeeman term is optional because it changes the phase diagram qualitatively: when $g_Z b \neq 0$, it acts as a linear source for the opposite component and removes the exact degeneracy of the two chiralities. Comparisons with the phase boundary of Ref. [6] must use $g_Z = 0$; nonzero values instead implement the microscopic extension of Ref. [7].

The model has equal component stiffnesses, no mixed-gradient term, and fixed quadratic coefficients. A nonuniform disorder profile is unsupported. Transport terminals are disabled; only natural order-parameter boundary conditions are supported.

4 s+s and s+is [8, 9, 10]

4.1 Free energy and order-parameter equations

Let ψ_1 and ψ_2 denote the two dimensionless isotropic s -wave components. Define

$$k_2 = k_{2/1} \equiv \text{k2_over_k1}, \quad k_{12} \equiv \text{mixed_gradient_k12}, \quad (48)$$

$$\gamma_J \equiv \text{josephson_gamma}, \quad \gamma_2 \equiv \text{phase_gamma2}, \quad \gamma_3 \equiv \text{density_gamma3}. \quad (49)$$

The two components share the solver disorder profile $\varepsilon(\mathbf{r}, t)$, but have independently tunable responses

$$a_{i,\text{eff}}(\mathbf{r}, t) = a_i + c_i[1 - \varepsilon(\mathbf{r}, t)], \quad c_1 \equiv \text{disorder_coupling1}, \quad c_2 \equiv \text{disorder_coupling2}. \quad (50)$$

A positive c_i therefore suppresses component i where $\varepsilon < 1$. With $k_1 = 1$, the implemented dimensionless free-energy density is

$$f = a_{1,\text{eff}}|\psi_1|^2 + a_{2,\text{eff}}|\psi_2|^2 + \frac{b_1}{2}|\psi_1|^4 + \frac{b_2}{2}|\psi_2|^4 \quad (51)$$

$$- \gamma_J(\psi_1\psi_2^* + \psi_2\psi_1^*) + \frac{\gamma_3}{2}|\psi_1|^2|\psi_2|^2 + \frac{\gamma_2}{2}(\psi_1^{*2}\psi_2^2 + \psi_2^{*2}\psi_1^2) \quad (52)$$

$$+ |\mathbf{D}\psi_1|^2 + k_2|\mathbf{D}\psi_2|^2 + k_{12}[(\mathbf{D}\psi_1)^* \cdot \mathbf{D}\psi_2 + (\mathbf{D}\psi_2)^* \cdot \mathbf{D}\psi_1]. \quad (53)$$

The first line and the bilinear term give the minimal two-band model of Refs. [8, 9]. The γ_2 , γ_3 , and k_{12} terms add the phase-sensitive quartic, density, and isotropic mixed-gradient couplings used for the $s + is$ model of Ref. [10].

The implemented dissipative extension is the negative gradient flow of Eq. (53):

$$r_1 D_t \psi_1 = (D_x^2 + D_y^2) \psi_1 + k_{12} (D_x^2 + D_y^2) \psi_2 - a_{1,\text{eff}} \psi_1 - b_1 |\psi_1|^2 \psi_1 - \frac{\gamma_3}{2} |\psi_2|^2 \psi_1 - \gamma_2 \psi_2^2 \psi_1^* + \gamma_J \psi_2, \quad (54)$$

$$r_2 D_t \psi_2 = k_2 (D_x^2 + D_y^2) \psi_2 + k_{12} (D_x^2 + D_y^2) \psi_1 - a_{2,\text{eff}} \psi_2 - b_2 |\psi_2|^2 \psi_2 - \frac{\gamma_3}{2} |\psi_1|^2 \psi_2 - \gamma_2 \psi_1^2 \psi_2^* + \gamma_J \psi_1. \quad (55)$$

The variational condensate current, transport current, and Poisson-normalized current are

$$\mathbf{J}_{\text{cond}} = \text{Im}(\psi_1^* \mathbf{D} \psi_1) + k_2 \text{Im}(\psi_2^* \mathbf{D} \psi_2) \quad (56)$$

$$+ k_{12} \text{Im}(\psi_1^* \mathbf{D} \psi_2 + \psi_2^* \mathbf{D} \psi_1), \quad (57)$$

$$\mathbf{J}_{\text{transport}} = \frac{\mathbf{J}_{\text{cond}}}{c_{\text{em}}}, \quad \mathbf{j}_s = \frac{\mathbf{J}_{\text{transport}}}{\beta_{\text{em}}}. \quad (58)$$

A positive γ_J minimizes the energy for equal component phases; a negative value favors a relative phase of π . When $\gamma_J = 0$, a positive γ_2 favors the time-reversal-related relative phases $\arg(\psi_2/\psi_1) = \pm\pi/2$ of an $s + is$ state.

With `include_screening=True`, local electromagnetic relaxation uses

$$\beta_{\text{em}} \partial_t \mathbf{A} = \mathbf{J}_{\text{cond}} - \kappa^2 (\partial_y B, -\partial_x B), \quad B = (\nabla \times \mathbf{A})_z. \quad (59)$$

The legacy arbitrary-mesh implementation evolves $B - H$ with the physical boundary condition $B = H$. Local screening requires $c_{\text{em}} = 1$ so that Eq. (59) is the variation of the same free energy as Eqs. (54)–(55). Prescribed-vector-potential simulations retain arbitrary positive c_{em} . In both modes, stored currents use Eq. (58); public physical currents restore the factor β_{em} .

The legacy arbitrary-mesh variational insulating-boundary conditions are coupled when $k_{12} \neq 0$:

$$\mathbf{n} \cdot \mathbf{D} \psi_1 + k_{12} \mathbf{n} \cdot \mathbf{D} \psi_2 = 0, \quad (60)$$

$$k_2 \mathbf{n} \cdot \mathbf{D} \psi_2 + k_{12} \mathbf{n} \cdot \mathbf{D} \psi_1 = 0. \quad (61)$$

4.2 Dimensional origin and SI units

The minimal Kogan–Schmalian coefficients have the microscopic clean-limit form below. A phenomenological SI functional that also includes the Ref. [10] couplings is

$$F_{\text{SI}} = \int dV \left\{ \sum_{\nu=1}^2 \left[a_{\nu}^{(G)} |\Delta_{\nu}|^2 + \frac{b_{\nu}^{(G)}}{2} |\Delta_{\nu}|^4 + K_{\nu}^{(G)} |\mathbf{\Pi} \Delta_{\nu}|^2 \right] - \gamma_J^{(G)} (\Delta_1 \Delta_2^* + \Delta_2 \Delta_1^*) \right. \quad (62)$$

$$+ \frac{\Gamma_3^{(G)}}{2} |\Delta_1|^2 |\Delta_2|^2 + \frac{\Gamma_2^{(G)}}{2} (\Delta_1^{*2} \Delta_2^2 + \Delta_2^{*2} \Delta_1^2) \quad (63)$$

$$\left. + K_{12}^{(G)} [(\mathbf{\Pi} \Delta_1)^* \cdot \mathbf{\Pi} \Delta_2 + \text{c.c.}] + \frac{B^2}{2\mu_0} \right\}, \quad (64)$$

where $\mathbf{\Pi} = \nabla + 2\pi i \mathbf{A}/\Phi_0$, $\Phi_0 = h/(2e)$, and Δ_{ν} is the dimensional band gap. Superscript (G) labels dimensional GL coefficients, not powers.

For the clean, isotropic weak-coupling model, the paper gives [9]

$$a_\nu^{(G)} = \frac{N(0)}{\eta_\lambda} (S_\nu - n_\nu \eta_\lambda \tau), \quad b_\nu^{(G)} = \frac{N(0)}{W^2} n_\nu, \quad (65)$$

$$\gamma_J^{(G)} = \frac{N(0)}{\eta_\lambda} \lambda_{12}, \quad K_\nu^{(G)} = \frac{N(0) \hbar^2 v_\nu^2}{6W^2} n_\nu, \quad (66)$$

$$\eta_\lambda = \lambda_{11} \lambda_{22} - \lambda_{12}^2, \quad S_1 = \lambda_{22} - n_1 \eta_\lambda S, \quad S_2 = \lambda_{11} - n_2 \eta_\lambda S, \quad (67)$$

$$W^2 = \frac{8\pi^2 (k_B T_c)^2}{7\zeta(3)}, \quad S = \ln \left(\frac{2e^{\gamma_E} \hbar \omega_D}{\pi k_B T_c} \right), \quad \tau = 1 - \frac{T}{T_c}. \quad (68)$$

Here $N(0)$ is the total single-spin density of states at the Fermi level; $n_\nu = N_\nu/N(0)$ the band fraction; $\lambda_{\nu\mu}$ the dimensionless pairing matrix; v_ν the Fermi velocity; ω_D the cutoff frequency; and γ_E Euler's constant. Equations (66) are one weak-coupling realization; Eq. (64) also applies phenomenologically.

SI units are: $\Delta_\nu, W, k_B T_c$ [J], $N(0)$ [$\text{J}^{-1}\text{m}^{-3}$], $a_\nu^{(G)}, \gamma_J^{(G)}$ [$\text{J}^{-1}\text{m}^{-3}$], $b_\nu^{(G)}, \Gamma_2^{(G)}, \Gamma_3^{(G)}$ [$\text{J}^{-3}\text{m}^{-3}$], $K_\nu^{(G)}, K_{12}^{(G)}$ [$\text{J}^{-1}\text{m}^{-1}$], v_ν [m s^{-1}], $\mathbf{A}$ [Tm], B [T], and F_{SI} [J]. $n_\nu, \lambda_{\nu\mu}, \eta_\lambda, S, \tau$ are dimensionless.

With gap scale Δ_0 , length ξ , and energy-density scale $f_0 = K_1^{(G)} \Delta_0^2 / \xi^2$, the code coefficients are

$$a_\nu = \frac{a_\nu^{(G)} \xi^2}{K_1^{(G)}}, \quad b_\nu = \frac{b_\nu^{(G)} \Delta_0^2 \xi^2}{K_1^{(G)}}, \quad (69)$$

$$k_{2/1} = \frac{K_2^{(G)}}{K_1^{(G)}}, \quad k_{12} = \frac{K_{12}^{(G)}}{K_1^{(G)}}, \quad (70)$$

$$\gamma_J = \frac{\gamma_J^{(G)} \xi^2}{K_1^{(G)}}, \quad \gamma_{2,3} = \frac{\Gamma_{2,3}^{(G)} \Delta_0^2 \xi^2}{K_1^{(G)}}. \quad (71)$$

If $a_{i,\text{eff}}^{(G)} = a_i^{(G)} + c_i^{(G)}(1 - \varepsilon)$, then $c_i = c_i^{(G)} \xi^2 / K_1^{(G)}$ in the same scaling. Other scales are equivalent if the whole functional is rescaled consistently.

4.3 Paper-aligned $s + is$ example

Ref. [10] does not provide a complete nondimensionalization into the repository's variables. Assuming the code's band-1 $K_1/\xi_1/\Delta_0$ scaling, its illustrative mass choice $m_1 = m_c = 1$, $m_2 = 2$ maps to

$$\begin{aligned} a_1 = a_2 = -1, \quad b_1 = b_2 = 1, \quad k_{2/1} = \frac{m_1}{m_2} = 0.5, \\ \gamma_J = 0, \quad \gamma_2 = 0.5, \quad \gamma_3 = 0, \quad k_{12} = \frac{m_1}{2m_c} = 0.5. \end{aligned} \quad (72)$$

These values favor the two $s + is$ relative phases $\pm\pi/2$. If one additionally assumes equal physical diffusion constants, the band-1 clock would give $r_2 = 0.5$; that is an extra dynamical assumption, not a value specified by Ref. [10].

4.4 Lin–Maiti–Chubukov defect comparison

The reproducibility driver `my_scripts/reproduce_lin_maiti_chubukov_prb94_064519.py` targets Figs. 3 and 4 of Ref. [10]. They are the most convenient tests of the distinction because both

use the same GL coefficients, require no vortex initial condition, and test complementary predictions: the weak-defect power law and the spatial symmetry of the induced field. In both models the script uses

$$\alpha_1 = \alpha_2 = -1, \quad \beta_1 = \beta_2 = 1, \quad \gamma_2 = 0.5, \quad \gamma_1 = \gamma_3 = 0, \quad m_1 = m_c = 1, \quad m_2 = 2. \quad (73)$$

The $s + is$ code map is Eq. (72), augmented by $c_1 = c_2 = 1$. With a defect indicator χ_D and strength $\tilde{\alpha}$, the common solver profile

$$\varepsilon = 1 - \tilde{\alpha}\chi_D \quad (74)$$

then gives $a_{i,\text{eff}} = -1 + \tilde{\alpha}\chi_D$, exactly the paper's quadratic-coefficient defect.

The repository stores the $s + d$ fields as $\psi_1 = \psi_s$ and $\psi_2 = \psi_d$ and uses the d -sector stiffness and coherence length as its units. The corresponding $s + id$ parameters are

$$\eta_s = 2, \quad \eta_v = -1, \quad \nu = 1, \quad \tau_1 = 1, \quad \tau_3 = 0, \quad \tau_4 = 0.5, \quad c_\nu = 1. \quad (75)$$

The negative η_v converts the code's y -minus- x directional convention to the paper's x -minus- y convention. Equation (74) makes both $s + id$ quadratic drives equal to $1 - \tilde{\alpha}\chi_D$. Since $\xi_1/\xi_d = \sqrt{m_2/m_1} = \sqrt{2}$, comparison in the paper's units requires

$$\mathbf{r}_{\text{code}} = \sqrt{2} \frac{\mathbf{r}_{\text{paper}}}{\xi_1}, \quad \frac{B_{\text{paper}}}{H_{c2,1}} = 2 \frac{B_{\text{code}}}{B_{c2,d}}. \quad (76)$$

Both models are initialized in one homogeneous chiral minimum, $\psi_1 = \sqrt{2}$ and $\psi_2 = i\sqrt{2}$; changing the chirality reverses the field without changing the reported magnitude.

For the Fig. 3 analogue, the square-defect indicator is

$$\chi_D = \Theta(r_0 - |x|)\Theta(r_0 - |y|), \quad r_0 = 1.5, \quad (77)$$

and the script fits $\max|B - B_0| = C\tilde{\alpha}^p$ on a logarithmic strength scan. The expected signatures are $p = 2$ for $s + is$ and $p = 1$ for $s + id$. For the Fig. 4 analogue,

$$\chi_D = \Theta(r_0 - |x|)\Theta(r_0 - |y|), \quad r_0 = 1, \quad \tilde{\alpha} = 0.5, \quad (78)$$

which yields the fourfold corner pattern of $s + is$ and the twofold pattern of $s + id$. Each field-map panel uses its own symmetric color range because the absolute amplitudes can differ by orders of magnitude.

The driver uses a D_4 -symmetric triangular mesh, temporal gauge, local screening, and subtraction of the identically discretized zero-defect field B_0 . It takes $\beta_{\text{em}} = r_1 = r_2 = r_s = 1$ as a numerical relaxation convention; these kinetic choices do not change the static extrema. Ref. [10] does not specify the finite domain, mesh, boundary conditions, diffusion constants, conductivity, or penetration depth. The script therefore takes $\lambda/\xi_1 = 4$ by default (and $\kappa_d = 4\sqrt{2}$ for $s + id$) and presents a finite-domain reproduction of the symmetry and scaling laws rather than a pixel-level prediction of the absolute field. The three resolution presets can be run as

```
SCRIPT=my_scripts/reproduce_lin_maiti_chubukov_prb94_064519.py
python "$SCRIPT" --preset smoke
python "$SCRIPT" --preset quick
python "$SCRIPT" --preset paper
```

Existing HDF5 checkpoints are reused; `--plot-only` redraws their figures and `--force` recomputes the selected cases.

4.5 Parameters

Code name	Meaning	Default
a_1 (a1), a_2 (a2)	Dimensionless $a_\nu^{(G)}$; contain a constant interband contribution and a term proportional to $1 - T/T_c$.	-1, -1
b_1 (b1), b_2 (b2)	Dimensionless quartic self-interactions corresponding to $b_\nu^{(G)} = N(0)n_\nu/W^2$.	1, 1
$k_{2/1}$ (k2_over_k1)	Stiffness ratio $K_2^{(G)}/K_1^{(G)}$; depends on band fractions and squared Fermi velocities.	1
k_{12} (mixed_gradient_k12)	Isotropic intercomponent gradient coupling $K_{12}^{(G)}/K_1^{(G)}$.	0
γ_J (josephson_gamma)	Dimensionless bilinear Josephson coupling corresponding to $\gamma_J^{(G)} = N(0)\lambda_{12}/\eta_\lambda$ in the minimal clean model.	0
γ_2 (phase_gamma2)	Phase-sensitive quartic coupling; positive values favor relative phase $\pm\pi/2$ when the bilinear term vanishes.	0
γ_3 (density_gamma3)	Intercomponent density-density quartic coupling.	0
c_1, c_2 (disorder_coupling1, disorder_coupling2)	Responses of the two quadratic coefficients to the shared profile in Eq. (50).	0, 0
r_1 (relaxation1), r_2 (relaxation2)	Phenomenological TDGL kinetic coefficients; absent from the static Kogan–Schmalian functional.	1, 1
c_{em} (em_coupling)	Condensate-to-transport current conversion. Local screening requires one for variational consistency.	1
β_{em} (beta_em)	Electromagnetic relaxation and stored-current normalization in Eqs. (58)–(59).	1

The validation conditions are

$$b_1 > 0, \quad b_2 \geq 0, \quad k_2 > 0, \quad k_{12}^2 < k_2, \quad (79)$$

$$r_1 > 0, \quad r_2 > 0, \quad c_{\text{em}} > 0, \quad \beta_{\text{em}} > 0. \quad (80)$$

For $b_2 > 0$, quartic coercivity additionally requires

$$\frac{\gamma_3}{2} - |\gamma_2| > -\sqrt{b_1 b_2}. \quad (81)$$

For the passive-band case $b_2 = 0$, the stricter conditions are $a_2 > 0$ and $\gamma_3/2 - |\gamma_2| \geq 0$. a_1, a_2 may have either sign; an uncoupled uniform band with $a_n < 0$ has $|\psi_n| = \sqrt{-a_n/b_n}$.

4.6 Limitations and approximations

The microscopic coefficient map in Ref. [9] is derived for a clean, isotropic, weak-coupling material, not the dirty limit. The GL result that the two gaps are proportional near the common T_c is more general and also has dirty-limit derivations, but those dirty microscopic coefficients are not encoded here. The relaxation coefficients r_1, r_2 remain phenomenological.

Equation (50) is a phenomenological, linear response of both quadratic coefficients to one shared scalar profile. With $c_1 = c_2 = 1$ and Eq. (74), it does reproduce the phenomenological $\alpha_i \mapsto \alpha_i + \tilde{\alpha}\chi_D$ defect used by Ref. [10]. It does not represent a microscopic or multiorbital impurity potential, nor does it encode impurity-induced interband scattering. The new quartic and mixed-gradient coefficients are independent phenomenological inputs unless a consistent microscopic mapping is supplied.

Near T_c , the two stored fields do not imply two independent GL coherence lengths. The relaxation coefficients remain phenomenological, and the local screening equation is a dissipative equilibrium-finding dynamics rather than a material-specific electrodynamic model. Scalar terminal order-parameter values remain unsupported because their two-component mapping is unspecified.

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